Opening & Reading Files

`open(path, mode)` connects Python to a file. Always close it, or use `with`.

1) Read the whole file

Example :

```
path = "Students_record.txt"
f = open(path, "r", encoding="utf-8")
print(f.read(60))
f.close()
```

Output :

```
STUDENT RECORDS
--------------------
Student ID: 101
Name: R
```

read() with no number reads everything, read(60) reads the first 60 characters

☆ RAW STRINGS

- On Windows write `r"D:\folder\file.txt"`. The `r` keeps every backslash literal, so `\n` and `\t` are not escapes.

2) Read line by line

Example :

```
f = open("Students_record.txt", "r", encoding="utf-8")
print(f.readline(), end="")
print(f.readline(), end="")
print(f.readline(5))
f.close()
```

Output :

```
STUDENT RECORDS
--------------------
Stude
```

each readline() continues where the last one stopped, a number limits the characters

3) Read all lines into a list, or loop

Example :

```
with open("Students_record.txt", "r") as f:
    lines = f.readlines()
print(lines[0])
print(len(lines))
count = 0
for line in lines:
    if line.startswith("Student ID:"):
        count += 1
print("Total student records:", count)
```

Output :

```
STUDENT RECORDS

16
Total student records: 3
```

WRITING, APPENDING & WITH

"w" replaces a file, "a" adds to the end. with closes the file for you.

1) File modes

Mode	Meaning	What happens
"r"	read	opens an existing file
"w"	write	creates the file or erases old content
"a"	append	adds content at the end
"x"	create	creates only if the file does not exist
"r+"	read + write	no erase on open
"w+"	write + read	erases existing content
"a+"	append + read	writes go to the end

2) Write (w) then append (a)

Example :

```
with open("Duplicate.txt", "w", encoding="utf-8") as f:
    f.write("Students Record Information\n")
    f.write("Name : Sanatan Khanolkar\n")
with open("Duplicate.txt", "a", encoding="utf-8") as f:
    f.write("Subject : Data Science\n")
with open("Duplicate.txt", "r", encoding="utf-8") as f:
    print(f.read())
```

Output :

```
Students Record Information
Name : Sanatan Khanolkar
Subject : Data Science
```

\n starts a new line in the file. Practise on a duplicate file, never your only copy

3) Why use with?

Example :

```
with open("Duplicate.txt", "r", encoding="utf-8") as f:
    print("Inside block:", f.closed)
print("Outside block:", f.closed)
```

Output :

```
Inside block: False
Outside block: True
```

with closes the file when the block ends, even if an error happens inside it

R+, A+, X & SEEK()

Read/write modes share one file position. `seek(0)` jumps to the start; `tell()` reports it.

1) r+ with seek()

Example :

```
with open("rw.txt", "w", encoding="utf-8") as f:
    f.write("ABCDEF")
with open("rw.txt", "r+", encoding="utf-8") as f:
    f.seek(0)
    f.write("XYZ")
    f.seek(0)
    print(f.read())
```

Output :

```
XYZDEF
```

writing overwrites characters at the current position, seek(0) goes back to read them

2) a+ appends, then reads

Example :

```
with open("student.txt", "w", encoding="utf-8") as f:
    f.write("Name: Neha Joshi\n")
with open("student.txt", "a+", encoding="utf-8") as f:
    f.write("Course: Data Science\n")
    f.seek(0)
    print(f.read())
```

Output :

```
Name: Neha Joshi
Course: Data Science
```

3) x creates safely

Example :

```
import os
if os.path.exists("new_student.txt"):
    os.remove("new_student.txt")
for attempt in range(2):
    try:
        with open("new_student.txt", "x") as f:
            f.write("Name: Neha Joshi\n")
        print("New file created!")
    except FileExistsError:
        print("File exists. Not overwritten.")
```

Output :

```
New file created!
File exists. Not overwritten.
```

CSV FILES & PRACTICE

CSV stores rows and columns of text. Use `csv.writer` and `csv.reader`, with `newline=""`.

1) Create, append and read a CSV

Example :

```
import csv
with open("Students.csv", "w", newline="", encoding="utf-8") as f:
    writer = csv.writer(f)
    writer.writerow(["Name", "Course", "Marks"])
    writer.writerow(["Rahul", "Python", 85])
    writer.writerow(["Priya", "Data Science", 92])
with open("Students.csv", "a", newline="", encoding="utf-8") as f:
    csv.writer(f).writerow(["Kavya", "Python", 90])
with open("Students.csv", "r", newline="", encoding="utf-8") as f:
    for row in csv.reader(f):
        print(row)
```

Output :

```
['Name', 'Course', 'Marks']
['Rahul', 'Python', '85']
['Priya', 'Data Science', '92']
['Kavya', 'Python', '90']
```

every value read from a CSV is a string: use int(row[2]) before calculating

2) Practice tasks

Task 1 Shopping list: write, then read.

Solution :

```
with open("Shopping_List.txt", "w", encoding="utf-8") as f:
    f.write("Milk\nBread\nApples\n")
with open("Shopping_List.txt", "r", encoding="utf-8") as f:
    print(f.read())
```

```
Milk
Bread
Apples
```

Task 2 Append Rice, then count items.

Solution :

```
with open("Shopping_List.txt", "a", encoding="utf-8") as f:
    f.write("Rice\n")
count = 0
with open("Shopping_List.txt", "r", encoding="utf-8") as f:
    for line in f:
        if line.strip():
            count += 1
print("Number of shopping items:", count)
```

```
Number of shopping items: 4
```

Task 3 Search for a student name.

Solution :

```
name = input("Enter a student's name: ")
with open("Students_record.txt") as f:
    content = f.read()
if name.lower() in content.lower():
    print("Found")
else:
    print("Not found")
```

```
Enter a student's name: Priya Patil
Found
```